

GRA4159 Trends, Cycles, and Signal Extraction from a Macroeconomic Perspective

Growth Accounting and Total Factor Productivity The Production Function Approach

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Motivation and Basic Idea

- Economic growth can arise from:
 - Accumulation of inputs (capital and labor)
 - Improvements in efficiency or technology
- Growth accounting provides a framework to:
 - Decompose output growth into factor contributions
 - Measure productivity growth as a residual
- The approach relies on a neoclassical production function

Production Function

Assume aggregate output is produced according to a Cobb–Douglas production function:

$$Y_t = A_t K_t^\alpha L_t^{1-\alpha}$$

where:

- Y_t = real output (value added)
- K_t = capital input
- L_t = labor input
- A_t = total factor productivity (TFP)
- $\alpha \in (0, 1)$ = capital income share

Assumptions:

- Constant returns to scale
- Competitive factor markets

From Levels to Growth Rates

Taking logs and first differences:

$$\Delta \ln Y_t = \Delta \ln A_t + \alpha \Delta \ln K_t + (1 - \alpha) \Delta \ln L_t$$

This implies:

- Output growth is the sum of:
 - Capital input growth (weighted by α)
 - Labor input growth (weighted by $1 - \alpha$)
 - TFP growth
- TFP growth is measured as a residual

Measuring Total Factor Productivity

Rearranging:

$$\Delta \ln A_t = \Delta \ln Y_t - \alpha \Delta \ln K_t - (1 - \alpha) \Delta \ln L_t$$

Interpretation:

- TFP captures changes in:
 - Technology and innovation
 - Organization and management
 - Efficiency and utilization
- It also absorbs measurement error and cyclical effects

Practical Considerations

- Output: use real value added in constant prices
- Labor input: preferably hours worked
- Capital input:
 - Capital stock (via perpetual inventory method), or
 - Capital services (preferred)
- Factor shares:
 - Estimated from national accounts
 - Often allowed to vary over time
- TFP is best interpreted over medium- to long-run horizons